

1. Convert Celsius to Fahrenheit

Source code:

```
function celsiusToFahrenheit(celsius) {  
  return (celsius * 9 / 5) + 32;  
}  
  
let celsius = 25;  
let fahrenheit = celsiusToFahrenheit(celsius);  
  
console.log("Celsius:", celsius);  
console.log("Fahrenheit:", fahrenheit);
```

Output:

```
Problems Output Debug Console Terminal Ports  
[Running] node "c:\Users\Syed Sadi\Desktop\sqa.java\assingment.js"  
Celsius: 25  
Fahrenheit: 77  
  
[Done] exited with code=0 in 0.124 seconds
```

Explanation:

The function `celsiusToFahrenheit()` takes a temperature in Celsius and applies the formula.

2. Find the Factorial of a Number

Source Code:

```
function factorial(number) {  
    let result = 1;  
  
    for (let i = 1; i <= number; i++) {  
        result = result * i;  
    }  
  
    return result;  
}  
  
let number = 5;  
let result = factorial(number);  
  
console.log("Number:", number);  
console.log("Factorial:", result);
```

Output:

```
PS C:\Users\Syed Sadi\Desktop\sqa.java> node .\assingment.js  
Number: 5  
Factorial: 120  
PS C:\Users\Syed Sadi\Desktop\sqa.java>
```

Explanation:

function factorial(number) is used to calculate the factorial of a number.
for (let i = 1; i <= number; i++) { result = result * i; } calculate the factorial.

3. Check for Palindrome

Source Code:

```
function isPalindrome(word) {  
    let reversedWord = word.split("").reverse().join("");  
  
    return word === reversedWord;  
}  
  
let word = "madam";  
let result = isPalindrome(word);  
  
console.log("Word:", word);  
console.log("Is Palindrome:", result);
```

Output:

```
PS C:\Users\Syed Sadi\Desktop\sqa.java> node .\assingment.js  
Word: madam  
Is Palindrome: true  
PS C:\Users\Syed Sadi\Desktop\sqa.java> 
```

Explanation:

split("") - converts the word into an array of characters

reverse() - reverses the characters

join("") converts them back into a string

The original word is compared with the reversed word using ===

If they are the same, the function returns true; otherwise, it returns false.

4. Sum of Array Elements

Source Code:

```
function sumArray(numbers) {  
    let sum = 0;  
  
    for (let i = 0; i < numbers.length; i++) {  
        sum = sum + numbers[i];  
    }  
  
    return sum;  
}  
  
let numbers = [10, 20, 30, 40, 50];  
let result = sumArray(numbers);  
  
console.log("Array:", numbers);  
console.log("Sum:", result);
```

Output:

```
PS C:\Users\Syed Sadi\Desktop\sqa.java> node .\assingment.js  
Array: [ 10, 20, 30, 40, 50 ]  
Sum: 150  
PS C:\Users\Syed Sadi\Desktop\sqa.java>
```

Explanation:

The function `sumArray(numbers)` takes array of numbers.

`for (let i = 0; i < numbers.length; i++) { sum = sum + numbers[i]; }` loop do the sum calculation of the array.

5. FizzBuzz (Basic)

Source Code:

```
for (let i = 1; i <= 15; i++) {  
  
    if (i % 3 === 0 && i % 5 === 0) {  
        console.log("FizzBuzz");  
    }  
    else if (i % 3 === 0) {  
        console.log("Fizz");  
    }  
    else if (i % 5 === 0) {  
        console.log("Buzz");  
    }  
    else {  
        console.log(i);  
    }  
}
```

Output:

```
● PS C:\Users\Syed Sadi\Desktop\sqa.java> node .\assingment.js  
1  
2  
Fizz  
4  
Buzz  
Fizz  
7  
8  
Fizz  
Buzz  
11  
Fizz  
13  
14  
FizzBuzz  
❖ PS C:\Users\Syed Sadi\Desktop\sqa.java>  
○ PS C:\Users\Syed Sadi\Desktop\sqa.java> █
```

Explanation:

For loop go through the number 1 to 15

If/else if condition check the condition & the % operator gives the remainder after division